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The Laplacian of The Integral Of The Logarithmic Derivative of the Riemann-Siegel-Hardy Z-function

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Abstract

The integral $R(t) = \pi^{-1} \int \frac{\mathrm{d}}{\mathrm{d}t} \mathrm{lnZ}(t)\mathrm{d}t = \pi^{-1} \left(\ln \zeta \left(\frac{1}{2} + it\right) + i\vartheta(t)\right)$ of the logarithmic derivative of the Hardy Z function $Z(t) = e^{i\vartheta(t)}\zeta\left(\frac{1}{2} + it\right)$, where $\vartheta(t)$ is the Riemann-Siegel theta function, and $\zeta(t)$ is the Riemann zeta function, is used as a basis for the construction of a pair of transcendental entire functions $\nu(t) = -\nu(1-t) = -\left(\Delta R\left(\frac{i}{2} - it\right)\right)^{-1} = -G\left(\frac{i}{2} - it\right)$ where $G = -(\Delta R(t))^{-1}$ is the derivative of the additive inverse of the reciprocal of the Laplacian $\Delta f(t) = \ddot{f}(t)$ of $R(t)$ and $\chi(t) = -\chi(1-t) = \dot{\nu}(t) = -iH\left(\frac{i}{2} - it\right)$ where $H(t)$=$\dot{G}(t)$ has roots at the local minima and maxima of $G(t)$. When $H(t) = 0$ and $\dot{H}(t) = \ddot{G}(t) = \Delta G(t) > 0$, the point t marks a minimum of $G(t)$ where it coincides with a Riemann zero, i.e., $\zeta\left(\frac{1}{2} + it\right) = 0$, otherwise when $H(t) = 0$ and $\dot{H}(t) = \Delta G(t) < 0$, the point t marks a local maximum $G(t)$, marking midway points between consecutive minima. Considered as a sequence of distributions or wave functions, $\nu_n(t) = \nu(1 + 2n + 2t)$ converges to $\nu_\infty(t) = \lim_{n\to\infty} \nu_n(t) = \sin^2(\pi t)$ and $\chi_n(t) = \chi(1 + 2n + 2t)$ to $\chi_\infty(t) = \lim_{n\to\infty} \chi_n(t) = -8\cos(\pi t)\sin(\pi t)$

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1 Derivations

1.1 Standard Definitions

Let $\zeta(t)$ be the Riemann zeta function

$$\begin{aligned} \zeta(t) &= \textstyle\sum_{n=1}^{\infty} n^{-s} && \forall \operatorname{Re}(s) > 1 \\ &= (1-2^{1-s}) \textstyle\sum_{n=1}^{\infty} n^{-s}(-1)^{n-1} && \forall \operatorname{Re}(s) > 0 \end{aligned} \tag{1}$$

and $\vartheta(t)$ be Riemann-Siegel vartheta function $\vartheta(t)$

$$\vartheta(t) = -\frac{i}{2}\left(\ln\Gamma\left(\frac{1}{4}+\frac{it}{2}\right) - \ln\Gamma\left(\frac{1}{4}-\frac{it}{2}\right)\right) - \frac{\ln(\pi)t}{2} \tag{2}$$

where $\arg(z) = \frac{\ln(z)-\ln(\bar{z})}{2i}$ and $\overline{\Gamma(z)} = \Gamma(\bar{z})$. The Hardy Z function[Ivi13] can then be written as

$$Z(t) = e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right) \tag{3}$$

which can be mapped isometrically back to the ζ function

$$\zeta(t) = e^{-i\vartheta\left(\frac{i}{2}-it\right)} Z\left(\frac{i}{2}-it\right) \tag{4}$$

due to the isometry

$$t = \frac{1}{2} + i\left(\frac{i}{2}-it\right) \tag{5}$$

of the Mobius transforms[1] $f(t) = \frac{at+b}{ct+d} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with

$$\begin{pmatrix} -i & \frac{i}{2} \\ 0 & 1 \end{pmatrix} \text{and its inverse} \begin{pmatrix} i & \frac{1}{2} \\ 0 & 1 \end{pmatrix} \tag{6}$$

making possible the Riemann-Siegel-Hardy correspondence. Furthermore, let $S(t)$ be argument of ζ normalized by π defined by

$$\begin{aligned} S(t) &= \pi^{-1}\arg\left(\zeta\left(\tfrac{1}{2}+it\right)\right) \\ &= -\tfrac{i}{2\pi}\left(\ln\zeta\left(\tfrac{1}{2}+it\right) - \ln\zeta\left(\tfrac{1}{2}-it\right)\right) \end{aligned} \tag{7}$$

The Bäcklund counting formula gives the exact number of zeros on the critical strip up to level t, not just on the criticial line,

$$N(t) = \operatorname{Im}(R(t)) = \frac{\vartheta(t)}{\pi} + 1 + S(t) \tag{8}$$

The relationship between the functions $N(t)$, $S(t)$, and $Z(t)$ is demonstrated by

$$\ln\zeta\left(\frac{1}{2}+it\right) = \ln|Z(t)| + i\pi S(t) \tag{9}$$

These formulas are true independent of the Riemann hypothesis which posits that all complex zeros of $\zeta(s+it)$ have real part $s = \frac{1}{2}$. [Ivi13, Corrollary 1.8 p.13]

[1] Thanks to Matti Pitkänen for pointing out this is a Mobius transform pair, among other things

1.2 The Logarithmic Derivative of $Z(t)$ and its Integral

Let $Q(t)$ be the logarithmic derivative of $Z(t)$ given by

$$\begin{aligned} Q(t) &= \tfrac{\dot{\mathrm{Z}}(t)}{Z(t)} \\ &= \tfrac{\mathrm{d}}{\mathrm{d}t}\ln\mathrm{Z}(t) \\ &= i\left(\frac{\dot{\zeta}\left(\frac{1}{2}+it\right)}{\zeta\left(\frac{1}{2}+it\right)} + \frac{\Psi\left(\frac{1}{4}-\frac{i}{2t}\right)+\Psi\left(\frac{1}{4}+\frac{i}{2t}\right)-2\ln(\pi)}{4}\right) \\ &= i\left(\frac{\dot{z}(t)}{z(t)} + \frac{g^-(t)+g^+(t)-2\ln(\pi)}{4}\right) \end{aligned} \tag{10}$$

where

$$\Psi(x) = \frac{\mathrm{d}}{\mathrm{d}x}\ln(\Gamma(x)) = \frac{\frac{\mathrm{d}}{\mathrm{d}x}\Gamma(x)}{\Gamma(x)} \tag{11}$$

is the digamma function, the logarithmic derivative of the Γ function and $z(t)$ and $g^{\pm}(t)$ have been introduced to simplify the expressions. The function $Q(t)$ has singularities at $\pm\frac{i}{2}(4n-3)$ with residues

$$\operatorname*{Res}_{t=\frac{i}{2}(4n-3)}(Q(t)) = \frac{-8\zeta(2-2n)n^2+4\zeta(2-2n)n}{16\zeta(2-2n)n^2-8\zeta(2-2n)n} = -\frac{1}{2} \tag{12}$$

and

$$\operatorname*{Res}_{t=-\frac{i}{2}(4n-3)}(Q(t)) = \frac{-2\zeta(2n-1)+4\zeta(2n-1)n}{8\zeta(2n-1)n-4\zeta(2n-1)} = \frac{1}{2} \tag{13}$$

Now, the integral of the logarithmic derivative of Z is defined by

$$\begin{aligned} R(t) &= \pi^{-1}\int Q(t)\mathrm{d}t \\ &= \pi^{-1}\int \tfrac{\dot{\mathrm{Z}}(t)}{Z(t)}\mathrm{d}t \\ &= \pi^{-1}\int \tfrac{\mathrm{d}}{\mathrm{d}t}\ln\mathrm{Z}(t)\mathrm{d}t \\ &= \pi^{-1}\int i\left(\frac{\dot{z}(t)}{z(t)} + \frac{g^-(t)+g^+(t)-2\ln(\pi)}{4}\right)\mathrm{d}t \\ &= \pi^{-1}\int i\left(\frac{\dot{\zeta}\left(\frac{1}{2}+\mathrm{it}\right)}{\zeta\left(\frac{1}{2}+it\right)} + \frac{\Psi\left(\frac{1}{4}-\frac{i}{2t}\right)+\Psi\left(\frac{1}{4}+\frac{i}{2t}\right)-2\ln(\pi)}{4}\right)\mathrm{d}t \\ &= \pi^{-1}\left(\ln\zeta\left(\tfrac{1}{2}+it\right)+i\vartheta(t)\right) \\ &= \pi^{-1}\left(\ln\zeta\left(\tfrac{1}{2}+it\right)+\tfrac{i}{2}\left(i\left(\ln\Gamma\left(\tfrac{1}{4}-\tfrac{it}{2}\right)-\ln\Gamma\left(\tfrac{1}{4}+\tfrac{it}{2}\right)\right)-\ln(\pi)t\right)\right) \end{aligned} \tag{14}$$

1.3 The Laplacian

The Laplacian $= \nabla\cdot\nabla = \nabla^2$ is a differential operator which corresponds to the *divergence* of the *gradient* of a function $f(x)$ on a Euclidean space $x \in X \underline{\subseteq} \mathbb{R}^d$ and is denoted by

$$f(x) = \sum_{i=1}^{d}\frac{\partial}{\partial x_i^2}f(x_i)\forall x \in X \subseteq \mathbb{R}^d \tag{15}$$

and is simply the second derivative of a function when $d = 1$

$$f(x) = \ddot{f}(x) = \frac{\mathrm{d}}{\mathrm{d}x^2}f(x)\forall x \in X \subseteq \mathbb{R} \tag{16}$$

Let $G(t)$ be the additive inverse of the reciprocal(also known as multiplicative inverse) of the Laplacian of $R(t)$ interpreted as a partition function

$$\begin{aligned} G(t) &= -\tfrac{1}{\Delta R(t)} \\ &= \tfrac{8z(t)^2\pi}{z(t)^2(\dot{g}^-(t)-\dot{g}^+(t))-8(z(t)\ddot{z}(t)+\dot{z}(t)^2)} \end{aligned} \tag{17}$$

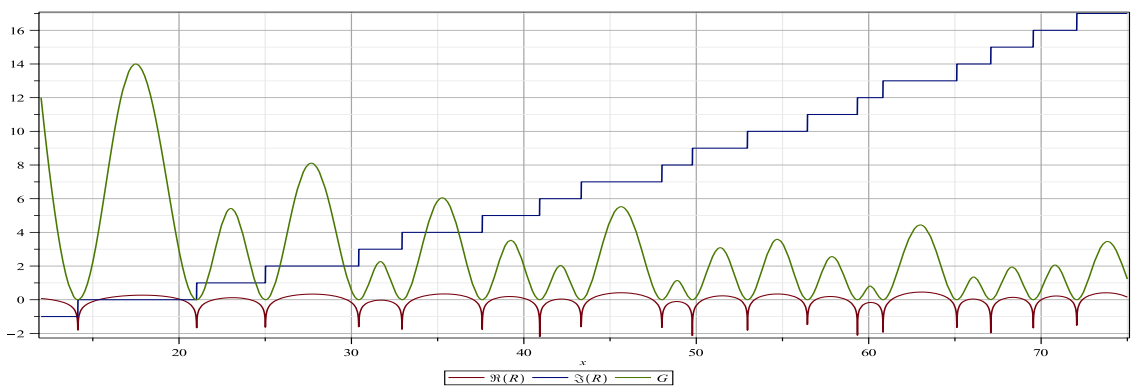

Figure 1: The Real and Imaginary Parts of $R(t)$ compared with $G(t)$

Now, let $H(t)$ be the derivative of $G(t)$ given by

$$\begin{aligned} H(t) &= \dot{G}(t) \\ &= \tfrac{\mathrm{d}}{\mathrm{d}t}\left(-\tfrac{1}{\Delta R(t)}\right) \\ &= -4i\pi z(t)\tfrac{z(t)^3\Delta g(t)+8(2z(t)^2\dddot{z}(t)-6z(t)\dot{z}(t)\ddot{z}(t)-4\dot{z}(t)^3)}{(z(t)^2+8(z(t)\ddot{z}(t)-\dot{z}(t)^2))^2} \end{aligned} \tag{18}$$

When $H(t) = 0$ and $\dot{H}(t) = \ddot{G}(t) = \Delta G(t) > 0$, the point t marks a minimum of $G(t)$ where it coincides with a Riemann zero, i.e., $\zeta\left(\frac{1}{2}+it\right) = 0$, otherwise when $H(t) = 0$ and $\dot{H}(t) = \Delta G(t) < 0$, the point t marks a local maximum $G(t)$, marking midway points between consecutive minima.

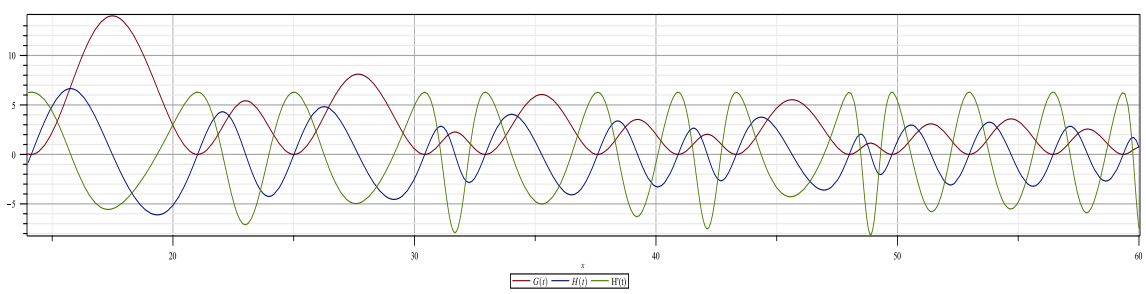

Figure 2: $G(t), H(t)$ and $\dot{H}(t)$

With the identity $t = \frac{1}{2}+i\left(\frac{i}{2}-it\right)$, define $\nu(t)$ as the Mobius transform of the partition function

$$\nu(t) = -\left(\Delta R\left(\frac{i}{2}-it\right)\right)^{-1} = -G\left(\frac{i}{2}-it\right) \tag{19}$$

then $\nu(t)$ has zeros at the positive odd integers, zero, and negative even integers. In the same way, let

$$\begin{aligned}\chi(t) &= \dot{\nu}(t) \\ &= -iH\left(\tfrac{i}{2} - it\right) \\ &= -4i\pi\zeta(t)\frac{\zeta(t)^3\left(\ddot{\Psi}\left(\frac{t}{2}\right)-\ddot{\Psi}\left(\frac{1}{2}-\frac{t}{2}\right)\right)-8(6\zeta(t)\dot{\zeta}(t)\ddot{\zeta}(t)+2\zeta(t)^2\dddot{\zeta}(t)+4\dot{\zeta}(t)^3)}{\zeta(t)^2\left(\left(\dot{\Psi}\left(\frac{t}{2}\right)-\dot{\Psi}\left(\frac{1}{2}-\frac{t}{2}\right)\right)+8(\zeta(t)\ddot{\zeta}(t)-\dot{\zeta}(t)^2)\right)^2}\end{aligned} \tag{20}$$

which also has zeros on the real line at the positive odd integers, zero, and the negative even integers.

$$\begin{aligned}\lim_{t\to 2n-1}\nu(t) &= \lim_{t\to 2n-1}\chi(t) &= 0\forall n>0 \\ \lim_{t\to 0}\nu(t) &= \lim_{t\to 0}\chi(t) &= 0 \\ \lim_{t\to -2n}\nu(t) &= \lim_{t\to -2n}\chi(t) &= 0\forall n<0\end{aligned} \tag{21}$$

Both $\nu(t)$ and $\chi(t)$ satisify similiar functional equations

$$\nu(t) = \nu(1-t) \tag{22}$$

and

$$\chi(t) = -\chi(1-t) \tag{23}$$

So the even $\mu(t)$ and odd $\psi(t)$ transcendental entire functions can be defined

$$\mu(t) = \nu\left(t+\frac{1}{2}\right) \tag{24}$$

$$\psi(t) = \chi\left(t+\frac{1}{2}\right) \tag{25}$$

which satisify the functional symmetries

$$\mu(t) = \mu(-t) \tag{26}$$

and

$$\psi(t) = -\psi(-t) \tag{27}$$

The function $\chi(t)$ has as a subset of its roots, the roots of the Rieman zeta function $\zeta(t)$, the converse is not true, since $\chi(t)$ is a function of $\zeta(t)$ and its first, second, and third derivatives.

$$\{t : \zeta(t) = 0\} \subset \{t : \chi(t) = 0\} \tag{28}$$

Let

$$\chi_n(t) = \chi(1+2n+2t) \tag{29}$$

and

$$\nu_n(t) = \nu(1+2n+2t) \tag{30}$$

which both satisify the symmetries

$$\chi_{-\frac{1}{2}}\left(\frac{t}{2}\right) = \chi(t) \tag{31}$$

$$\nu_{-\frac{1}{2}}\left(\frac{t}{2}\right) = \nu(t) \tag{32}$$

as well as

$$\chi_n\left(-\frac{1}{2}\right) = \chi(2n) \tag{33}$$

$$\nu_n\left(-\frac{1}{2}\right) = \nu(2n) \tag{34}$$

The sequence of wave functions $\nu_n(t)$ converges, thanks to quantum ergodicity, to

$$\nu_\infty(t) = \lim_{n\to\infty} \nu_n(t) = \sin^2(\pi t) \tag{35}$$

which has a limiting maximum

$$\begin{aligned} \lim_{n\to\infty} \max_{0<t<1} \nu_n(t) &= \max_{0<t<1} \nu_\infty(t) \\ &= \lim_{n\to\infty} \nu(2n) \\ &= \tfrac{8}{\pi} \\ &\cong 2.546479089470\ldots \end{aligned} \tag{36}$$

and the associated differential is

$$\begin{aligned} \chi_\infty(t) &= \lim_{n\to\infty} \chi_n(t) \\ &= \tfrac{4}{\pi}\tfrac{\mathrm{d}}{\mathrm{d}t}\nu_\infty(t) \\ &= \tfrac{4}{\pi}\tfrac{\mathrm{d}}{\mathrm{d}t}\sin^2(\pi t) \\ &= -8\sin(\pi t)\cos(\pi t) \end{aligned} \tag{37}$$

It is worth mentioning that the pair-correlation function for the zeros of $\zeta(s)$ is

$$r_2(x) = 1 - \frac{\sin^2(\pi x)^2}{\pi^2 x^2}$$

if the Riemann hypothesis is true. [11] The point $n = 1$ is where $\chi_n\left(-\frac{1}{2}\right)$ and $\nu_n\left(-\frac{1}{2}\right)$ attain their greatest values among the integers.

n	$\nu_n\left(-\frac{1}{2}\right)$	$\chi_n\left(-\frac{1}{2}\right)$
1	117.43532857805377782	9447.7593604718560
2	003.03320654562255410	0000.2816402783351
3	002.77176105375846239	0000.0589526080385
4	002.69653220844944185	0000.0240373109008
5	002.66095375057354766	0000.0132398305603
6	002.63970550787458574	0000.0088555925215
7	002.62532107269483772	0000.0060558883634
...	.....	.....
∞	$\frac{8}{\pi}$	0

Table 1: Even values of $\nu(n)$ and $\chi(n)$

This phenomena of having the first wave function having a much larger size than all of the remainders is mentioned in [Kna99, Theorem 22] The convergence of $\lim_{n\to\infty} \nu_n(t) \to \nu_\infty(t)$ and

$\lim_{n\to\infty} \chi_n(t) \to \chi_\infty(t)$ might be interpreted as a manifestation of quantum ergodicity [11, C p.16]. There is an essential singularity[CB90, 54 p.169] at $s_0 \cong 1.98757823\ldots$

$$\lim_{s\to s_0} \chi(s) = \infty \tag{38}$$

We also have the limits

$$\lim_{s\to\{0,1\}} \dot{\chi}(s) = 4\pi \tag{39}$$

The integral over this sine-squared kernel is

$$\int_0^1 \sin^2(\pi t)\mathrm{d}t = \frac{1}{2} \tag{40}$$

whereas

$$\int_0^1 \frac{\nu(t)}{\nu\left(\frac{1}{2}\right)}\mathrm{d}t \cong .46693755153559653755\ldots \tag{41}$$

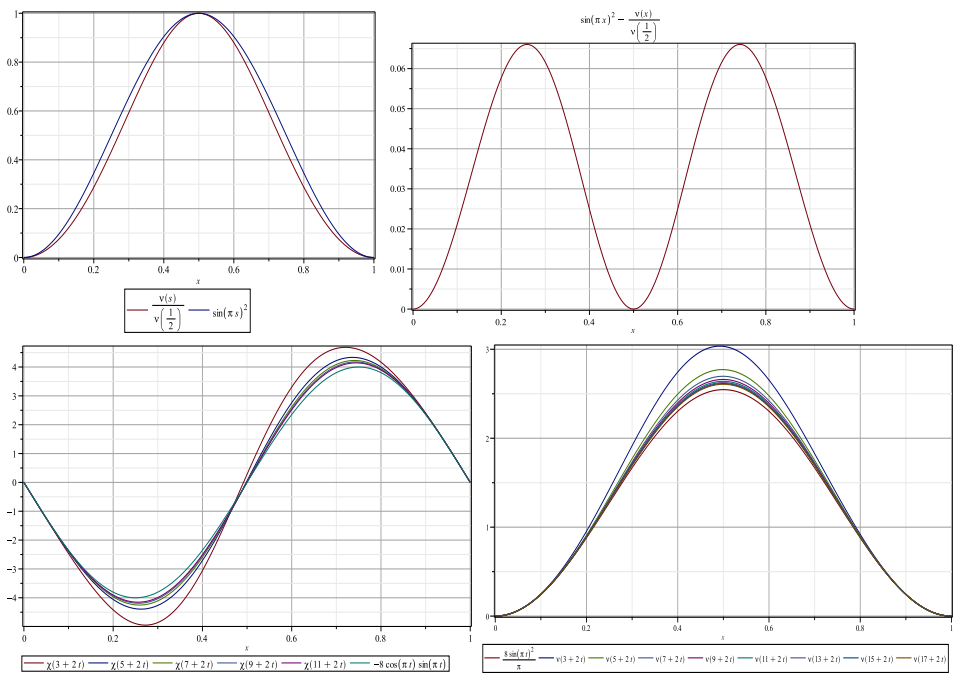

Figure 3: *Top Left*: The function $\nu(s)$ normalized by its maximum value on $[0,1]$ *Top Right*: The function $\nu(s)$ normalized by its maximum value and subtracted from $\sin^2(\pi x)$ *Bottom Left*: Convegence of $\chi_n(t) \to -8\cos(\pi t)\sin(\pi t)$ *Bottom Right*: Convegence of $\nu_n(t) \to \sin^2(\pi t)\frac{\pi}{8}$

2 Appendix

2.1 Wave Mechanics

2.2 The One-Dimensional Wave Equation

The one-dimensional wave equation is

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{x^2}u(x,t)+V(x,t)u(t,x)=-\frac{\hbar}{i}\frac{\partial}{\partial t}u(x,t) \tag{42}$$

[Flü94, II.A.4 p.25][Gan06, 4.2.1 p.][Coo03, Ch. 8 p.147]

2.3 The Poisson Bracket and Lagrangian Mechanics

"The" Poisson bracket, expressed with Einstein summation convention (for the repeated index i)

$$\{A,B\}\equiv\frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i}-\frac{\partial B}{\partial p_i}\frac{\partial A}{\partial q_i} \tag{43}$$

has the *antisymmetry* property

$$\{A,B\}=-\{B,A\} \tag{44}$$

and the so-called *Jacobi identity*

$$\{A,\{B,C\}\}-\{B,\{A,C\}\}+\{C,\{A,B\}\}=0 \tag{45}$$

Two quantitities A and B are said to *commute* if their Poisson bracket $\{A,B\}$ vanishes, that is, $\{A,B\}=0$. Hamilton's equations of motion for the system

$$\dot{p}_i=-\frac{H}{q_i} \tag{46}$$

$$\dot{q}_i=\frac{\partial H}{\partial q_i} \tag{47}$$

where H is a Legendre-transformed function of the Lagrangian called the Hamiltonian

$$H\equiv\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i-L(q_i,\dot{q}_i,t) \tag{48}$$

whose value for any given time t gives the *energy*

$$\mathcal{A}[q_i]=\int_{t_a}^{t_b}L(q_i,\dot{q}_i,t)\mathrm{d}t \tag{49}$$

of the system where $L(q_i,\dot{q}_i,t)$ is the Lagrangian of the system and

$$q_i(t)=q_i^{\mathrm{cl}}(t)+\delta q_i(t) \tag{50}$$

is an arbitrary path where $q_i^{\mathrm{cl}}(t)$ is the *classical orbit* or *classical path* of the system and

$$\delta q_i(t_a)\equiv q_i(t)-q_i^{\mathrm{cl}}(t) \tag{51}$$

[Kle04, 1.1]

2.3.1 The Euler-Lagrange equation

The Euler-Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i} \tag{52}$$

indicates that the action S given by

$$S = \int_{t_1}^{t_2} L(t)\mathrm{d}t \tag{53}$$

where

$$L(t) = T - V(x(t)) \tag{54}$$

is the *Lagrangian* and

$$T(t) = \frac{1}{2}m\dot{x}(t)^2 \tag{55}$$

is the *kinetic energy* which is stationary for the physical solutions $q_i(t)$.[Gan06, 4.2.1]

2.3.2 Quantum Mechanics of General Lagrangian Systems

The coordinate transformation

$$x^i = x^i(q^\mu) \tag{56}$$

implies the relation

$$\partial_\mu = \frac{\partial}{\partial q^\mu} = e^i_\mu(q)\partial_i \tag{57}$$

between the derivatives ∂_μ and

$$\partial_i \equiv \frac{\partial}{\partial x^i} \tag{58}$$

where

$$e^i_\mu(q) \equiv \partial_\mu x^i(q) \tag{59}$$

is a transformation matrix called the *basis p-ad* where p is the prefix corresponding to n, the dimension of x, monad when $n = 1$, dyad when $n = 2$, triad when n =3, and so on. Let

$$e^\mu_i(q) = \frac{\partial q^\mu}{\partial x^i} \tag{60}$$

be the inverse matrix called the *reciprocal p-ad*. The basis $p-$ ad and its recpiprocal satisify the orthogonality and completeness relations

$$e^i_\mu e^\nu_i = \delta^\nu_\mu \tag{61}$$

and

$$e^i_\mu e^\mu_j = \delta^i_j \tag{62}$$

The inverse of ∂_μ is

$$\partial_i = e^\mu_i(q)\partial_\mu \tag{63}$$

which is related to the curvilinear transform of the Cartesian quantum-mechanical momentum operators by

$$\hat{p}_i = -i\hbar\partial_i = -i\hbar e^\mu_i(q)\partial_\mu \tag{64}$$

The Hamiltonian operator for free particles is defined by

$$\hat{H}_0 = \hat{T} = \frac{1}{2M}\hat{p}^2 = -\frac{\hbar^2}{2M}\Delta \tag{65}$$

where its *metric tensor* is given by

$$g^{\mu\nu}(q) \equiv e_i^\mu(q)e_i^\nu(q) \tag{66}$$

and its inverse by

$$g_{\mu\nu}(q) \equiv e_\mu^i(q)e_\nu^i(q) \tag{67}$$

The Laplacian of a metric tensor is then expressed

$$\begin{aligned} \Delta &= \partial_i^2 \\ &= e^{i\mu}\partial_\mu e^{i\nu}\partial_\nu \\ &= e^{i\mu}e^{i\nu}\partial_\mu\partial_\nu + (e^{i\mu}\partial_\mu e^{i\nu})\partial_\nu \\ &= g^{\mu\nu}(q)\partial_\mu\partial_\nu - \Gamma_\mu^{\mu\nu}(q)\partial_\nu \end{aligned} \tag{68}$$

where

$$\Gamma_{\mu\nu}^\lambda(q) = -e_v^i(q)\partial_\mu e_i^\lambda(q) = e_i^\lambda(q)\partial_u e_\nu^i(q) \tag{69}$$

is the *affine-connection* [Kle04, 1.13][Pod28]

2.3.3 Noether's Theorem and Lie Groups

From Noether's theorem it is known that continuous symmetries have corresponding conservation laws.[Gan06, 4.2.1] Let

$$\alpha_s \tag{70}$$

be a continuous family of symmetries which is a 1-parameter subgroup $s \mapsto \alpha_s$ in the Lie group of symmetries. A Lie group is a group whose operations are compatible with the smooth structure. A smooth structure on a manifold allows for an unambiguous notion of smooth function.

2.4 Schroedinger's Time-Dependent Equation and Nonstationary Wave Motion

The function $\cos^2(t) = 1-\sin^2(t)$ arises in the Dirac phase averaving method of calculating transition probabilities of non-stationary states in the time-dependent Schroedinger equation. [Gol61, 15.1] 3

2.4.1 Operators and Observables: Dirac's Time-Dependent Theory

Let

$$y(x,t) = A\cos\left(\frac{2\pi px}{h}\right)\cos\left(\frac{2\pi\epsilon t}{h}\right)$$

where

$$\epsilon = hv \tag{71}$$

which relates the energies of a "matter wave" system, whatever that is, maybe a fermionic system, to the frequencies of quanta it emits or absorbs. There is always some arbitriness with choice of units. This wave function has the symmetry

$$\frac{\partial^2 y(x,t)}{x^2} = -\left(\frac{2\pi}{h}\right)^2 p^2 y(x,t) \tag{72}$$

so that the *kinetic energy* of a particle is obtained from the wave function

$$-\frac{(2\pi)^2}{2m}\frac{\partial^2 y(x,t)}{x^2} = \frac{p^2}{2m} y(x,t) \tag{73}$$

which means formally that

$$\frac{1}{2m}(p)^2 y(x,t) = -\frac{1}{2m}\left(\frac{2\pi}{i}\frac{\partial}{\partial x}\right)^2 \tag{74}$$

which suggests the identifiation of the *Schroedinger momentum operator*

$$\left(\frac{2\pi}{i}\frac{\partial}{\partial x}\right) \leftrightarrow p \tag{75}$$

The equation for "nonfree" particles is augmented by

$$\frac{m}{2}\left(\frac{2\pi}{i}\frac{\partial}{\partial x}\right)^2 y(x,t) = [\epsilon - V(x)]y(x,t) \tag{76}$$

where ϵ is the total energy of the particle and $V(x)$ is its potential energy. The first Schroedinger equation is then expressed

$$\epsilon y(x,t) = \left[\frac{1}{2m}\left(\frac{2\pi}{i}\frac{\partial}{\partial x}\right)^2 + V(x)\right] y(x,t) = \left[\frac{1}{2m}(p)^2 + V(x)\right] y(x,t) \tag{77}$$

or

$$H_{y(x,t)} = \epsilon y(x,t)$$

where H is the operator corresponding to the Hamiltonian of the (point) particle. [Gol61, 11.5] The *time-dependent Schroedinger equation* is written

$$\frac{\partial^2(x,t)}{\partial t^2} = -\frac{1}{(2\pi)^2}\epsilon^2 y(x,t) = -\left(\frac{H}{2\pi}\right)^2 y(x,t) \tag{78}$$

or equivalently as

$$0 = \left((2\pi)^2\frac{\partial^2}{\partial t^2} + H^2\right) y(x,t) = \left(H + i2\pi\frac{\partial}{\partial t}\right)\left(H - i2\pi\frac{\partial}{\partial t}\right) y(x,t) \tag{79}$$

where

$$Hy(x,t) = i2\pi\frac{\partial}{\partial t} y(x,t) \tag{80}$$

It is also worth mentioning that in Dirac's theory of the time-dependent Schroedinger there is a Hamiltonian of the form

$$H = H_0 + \lambda H_1(t) \tag{81}$$

which has a transition probability per unit time of

$$w(p \to -p) = \frac{V_0^2 \sin^2\left(2l\sqrt{\frac{2m\epsilon}{\hbar^2}}\right)}{2\sqrt{8m}\epsilon^{\frac{3}{2}} L}$$

where $V_0, \epsilon, h,$ and L are suitable constants. [Gol61, 15.5]

2.4.2 The String Theoretic Partition Function

The *string theoretic partition function* is defined as

$$Z_{st}(\omega) = Z_{st,R}(\omega) = \mathrm{Tr}[e^{2\pi i\omega_1 P - 2\pi\omega_2 H}] = (q\bar{q})^{-\frac{1}{24}} \mathrm{Tr}(q^{L_0^+} \bar{q}^{L_0^-}) \tag{82}$$

where L_0^+ and L_0^- are the Virasoro generators defined by

$$L_0^{\pm} = \frac{(p^+)^2 + (p^-)^2}{2} + \sum_{n=1}^{\infty} \alpha_{-n}^{\pm} \alpha_n^{\pm} \tag{83}$$

and $\alpha_{-n}^{\pm}\alpha_n$ are related to something called a Fubini-Veneziano field and p^+ is the *left*-momentum operator and p^- is *right-momentum* operator. [Lap08, 2.2.1]

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